

Philadelphia Flood Risk Ensemble Analysis Report

Case Study Area: Philadelphia County, PA (FIPS 42101)
Methodology: unsafe Flood Scenario Risk Ensemble Model
Report Generated: June 2026

This report compiles diagnostic visualizations and risk assessments generated by the unsafe flood scenario ensemble tool. By simulating physical flood risk under various uncertainties—including structure type, first-floor elevation (FFE), foundation type, and depth damage functions (DDFs)—this framework offers a comprehensive, probabilistic view of community vulnerability and expected damages.

1. Total Expected Annual Loss under Uncertainty

Comparing risk estimations with and without uncertainty reveals the significance of probabilistic assessment. Below is the total Expected Annual Loss (EAL) across the ensemble, compared against the standard deterministic estimate ('No Uncertainty') and segmented by two damage functions (HAZUS DDFs and NACCS DDFs).

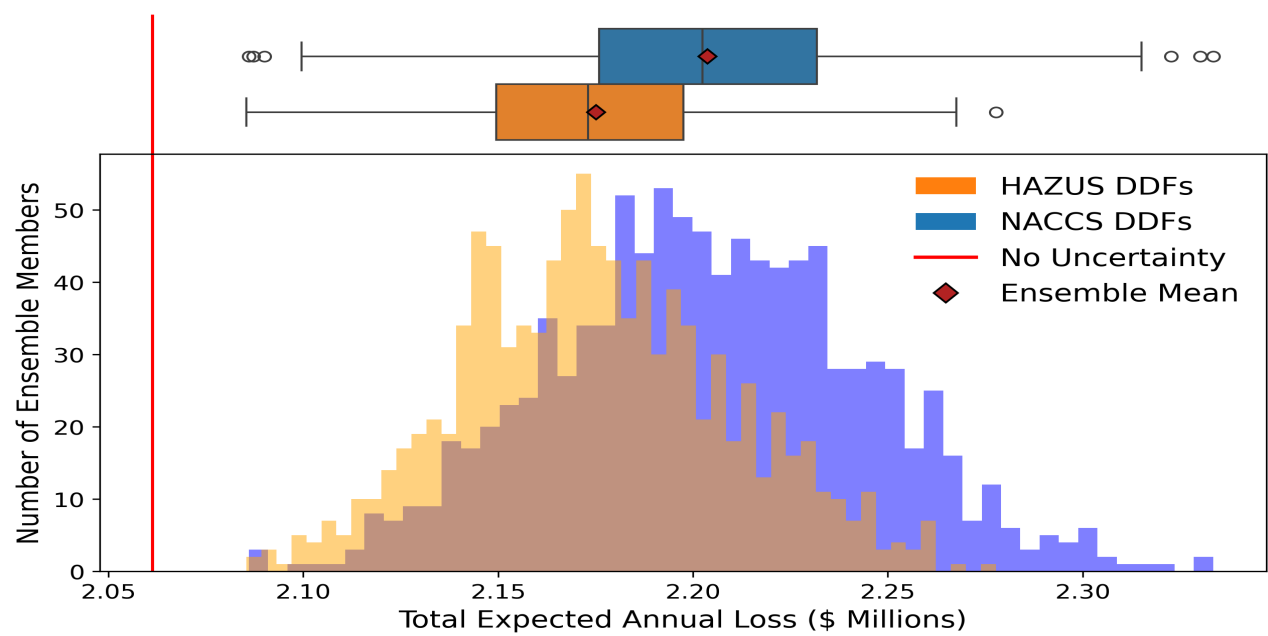


Figure 1: Distribution of total Expected Annual Losses across the ensemble compared with deterministic benchmark.

2. Single Property Risk Diagnostics

Even for the most at-risk properties, substantial variation in expected annual losses exists due to the interaction of uncertain structural attributes like foundation type and first-floor elevation (FFE). Below are diagnostic charts examining a single high-risk property across all simulated states of the world.

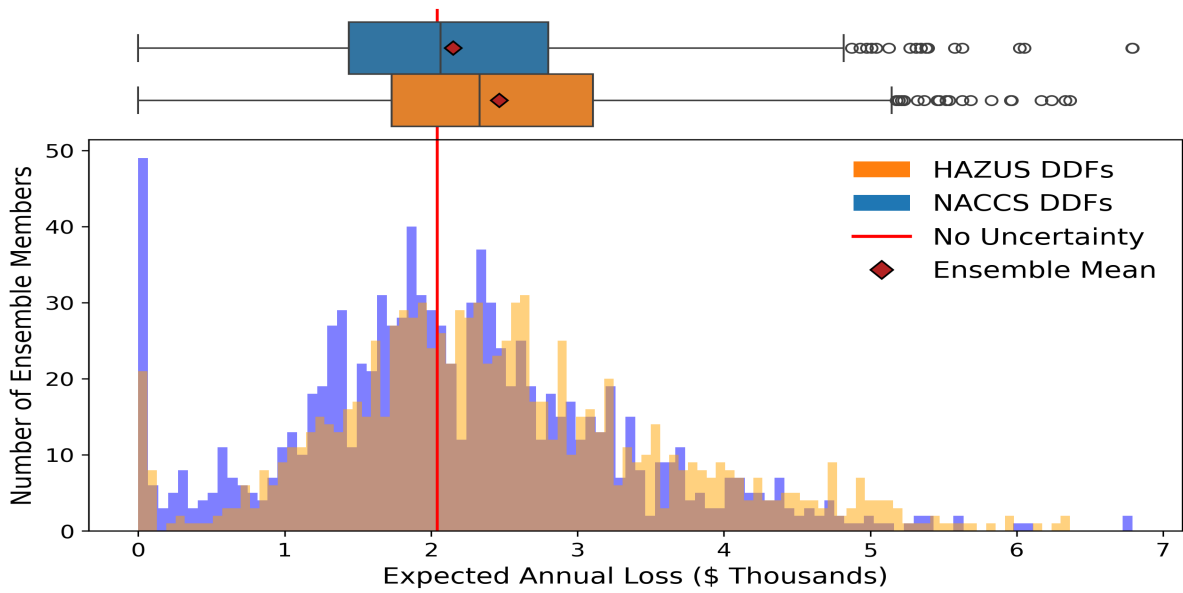


Figure 2: Expected Annual Loss distribution across SOWs for the highest-risk property.

Realizations of foundation type and FFE play a critical role in driving whether a property incurs damage during design flood scenarios. Higher first-floor elevations mitigate damages, but different foundation curves lead to distinct multimodal distributions.

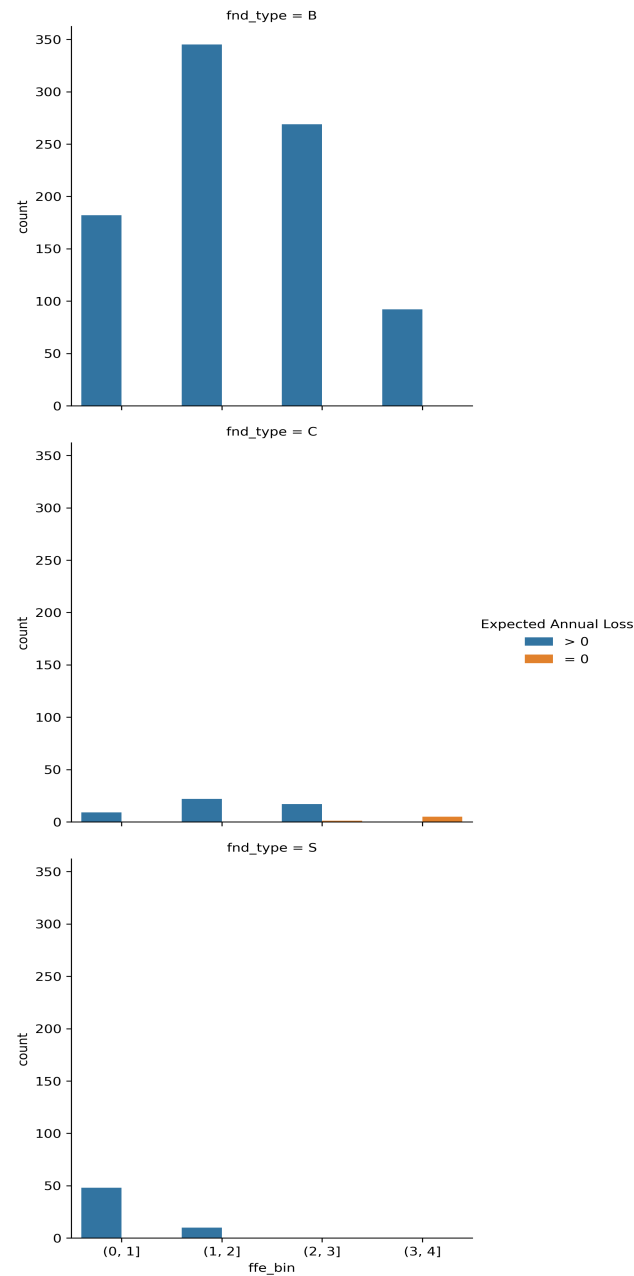


Figure 3: Count of realizations resulting in zero vs non-zero EAL across FFE and foundation types.

Structural Attributes and Expected Loss Interaction

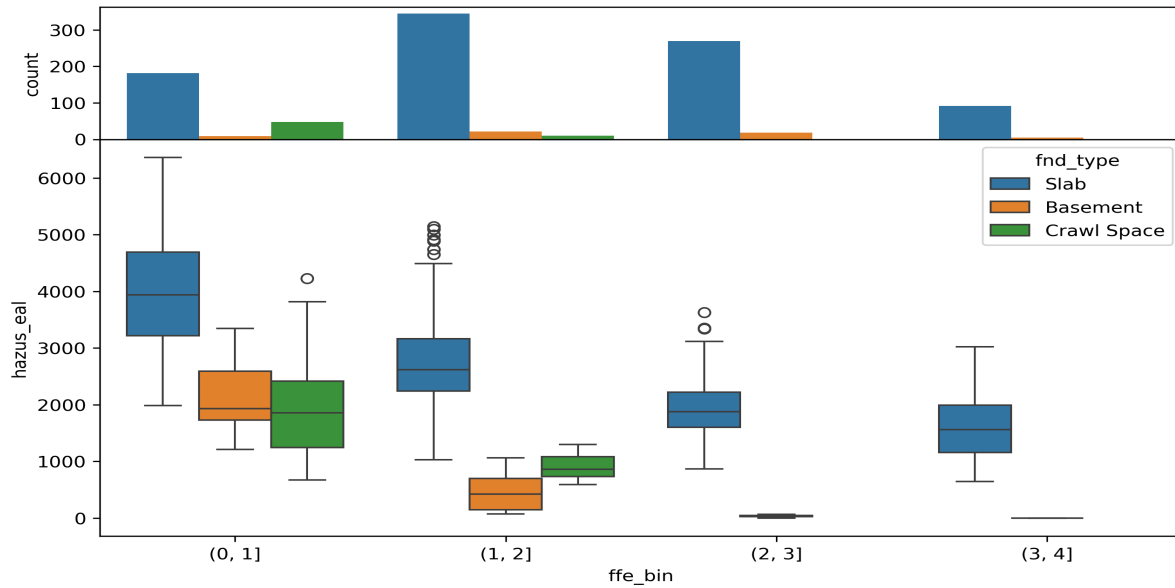


Figure 4: Expected Annual Loss boxplots and counts by FFE bin and foundation type.

3. Spatial Risk Assessment and Social Vulnerability

A key application of property-level risk assessment is overlaying spatial hazard data with social vulnerability indicators. Below, we overlay average structure-level Expected Annual Losses across the ensemble with disadvantaged census tracts designated by the Climate and Economic Justice Screening Tool (CEJST) in Philadelphia.

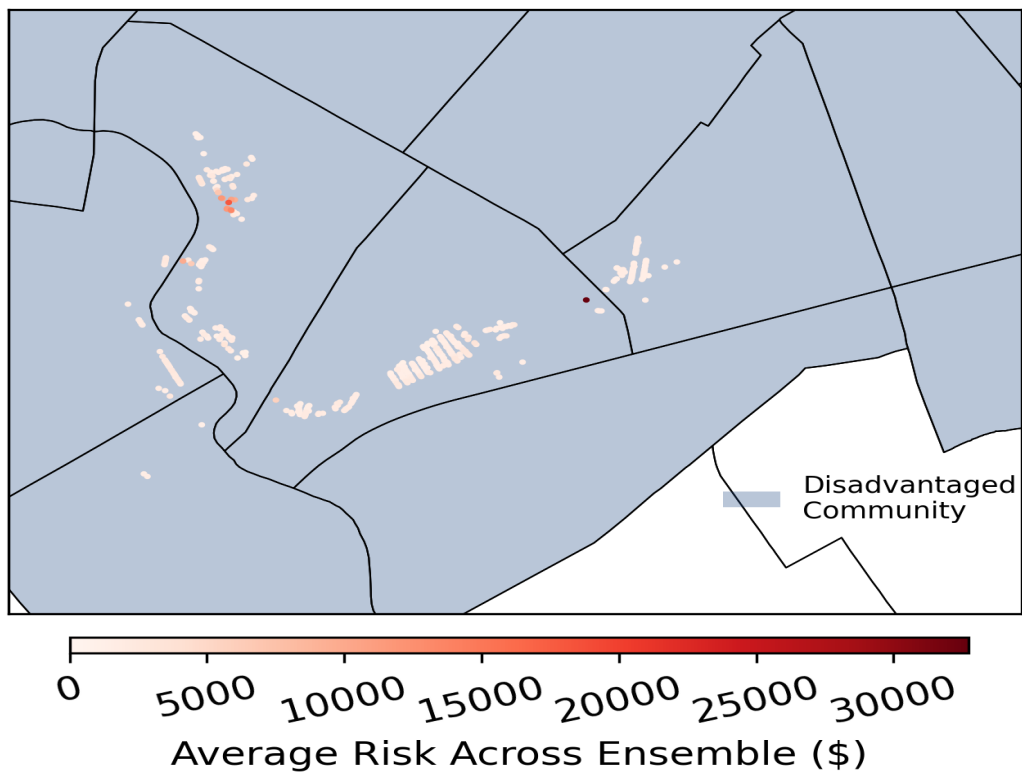


Figure 5: Property-level average flood risk overlaying CEJST disadvantaged communities.